

An artificial discontinuity in the tcpyPI pressure solver: a mathematical anatomy of issue #77

Analysis prepared from the tcpyPI source,
the PR #77 regression data, and the 2024 sample benchmark

July 12, 2026

Abstract

The potential-intensity solver in tcpyPI (a Python port of Emanuel’s p_{cm}) computes a central pressure by fixed-point iteration $x_{n+1} = g(x_n)$ and, for the sounding of PR #77 — Hurricane Francine’s pre-landfall environment in a counterfactual-SST sensitivity member — locks into a bit-exact period-2 cycle instead of converging. We show that the map g is *discontinuous*, that its only diagonal crossing falls inside a downward jump of 0.84 hPa at $x_d \simeq 950.8289947515$ hPa — so *no fixed point exists* — and we locate the source precisely: a discontinuous *selection rule* inside the CAPE routine (“integrate the signed buoyancy up to the topmost grid level where buoyancy is positive, then clamp at zero”). A one-bit sign decision at the 350 hPa level, taken on a marginal buoyancy feature worth $O(10^{-2}) \text{ J kg}^{-1}$, toggles the inclusion of 287 J kg^{-1} of negative area and moves the eyewall CAPE between 0 and 224 J kg^{-1} . Smooth toy models prove the pathology is a property of the selection functional, not of grid resolution, quadrature, or interpolation, and separate it from the *genuine* discontinuities of the underlying variational problem (the maximizer, by Berge’s maximum theorem; the reachable energy at barrier tangencies). The mechanism is generic — 20% of sampled real columns carry at least one jump of g — while the fatal alignment with the diagonal is rare ($\sim 0.04 \text{ K}$ of SST for this column), which is exactly why parameter sweeps find it. The repair is to define CAPE as the *value function* $\max_q W(q)$ of the running signed work — the convention that also matches the physics of forced eyewall ascent — and to solve the pressure equation with a residual-checked bracketed method. On both known failing soundings this converges immediately (exact roots 950.3137 and 960.7687 hPa); on a 7,093-column benchmark it changes no convergence flags and leaves 98.1% of intensities within 0.1 m s^{-1} , and it renders the period-2 “rescue” logic dead code.

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1 Conclusions first

1. **The failure is exactly reproducible and is not slow convergence.** The iteration alternates bit-exactly between $A = 950.653345443832$ and $B = 951.279007983853$ hPa; the separation (0.626 hPa) exceeds the 0.5 hPa stopping tolerance forever. Raising the iteration cap cannot help.
2. **No fixed point exists.** The map g jumps downward by 0.837 hPa at $x_d \simeq 950.829$ hPa; the residual $g(x) - x$ passes from +0.507 to -0.331 hPa across the jump without taking the value zero. A sign-change search reports x_d , but that is the location of a jump, not a root.
3. **The cause is one discontinuous selection.** In `cape()`, the level of neutral buoyancy is taken to be the *highest sampled level with positive buoyancy*. For the eyewall parcel this one-bit decision moves the terminal level between 750 and 350 hPa, dragging 287 J kg^{-1} of intervening negative area into a signed integral that is then clamped at zero: eyewall CAPE flips between 224.23 J kg^{-1} and 0. The saturated CAPE, outflow temperature, and outflow level are smooth there — the discontinuity is not in the outflow calculation and not a quadrature error. The rule is inherited verbatim from Emanuel’s `pc_min.m`: this is a defect of the reference algorithm itself, not of the Python port.
4. **The same mechanism explains the historical 1995 failure** (cycle $961.017 \leftrightarrow 961.671$ hPa, triggered by the 400 hPa sample; eyewall CAPE flips $0 \leftrightarrow 203 \text{ J kg}^{-1}$), while an adjacent-day control column has an ordinary continuous root and converges.
5. **The repair is one changed selection plus one changed solver.** Define CAPE as the maximum of the *same* running signed work over candidate terminal levels ($E_{\max} = \max_q W(q)$; Berge’s maximum theorem makes the value continuous even when the maximizing level jumps), and solve $g(x) = x$ with residual-checked bracketing. Negative-buoyancy layers still count against the work; nothing is skipped or free. Exact corrected results: $x_* = 950.3136889226$ hPa, $V_{\max} = 83.9065 \text{ m s}^{-1}$, $P_{\min} = 920.7517$ hPa for #77, and $x_* = 960.7687$ hPa, $V_{\max} = 74.4149 \text{ m s}^{-1}$ for the 1995 case, each confirmed by two independent implementations of the map. The upstream period-2 “rescue” (midpoint collapse) is then dead code; its midpoint ≈ 950.966 hPa was never a root of the stated equation.

2 The computation, stripped of meteorology

Glossary for the mathematician

Pressure as vertical coordinate. In atmospheric science the vertical coordinate is pressure p , measured in hectopascals (hPa); p decreases monotonically with altitude, from ~ 1000 hPa at sea level to 0 at the top of the atmosphere. “Above” means “smaller p ”. Data comes on a finite grid of *pressure levels* $p_0 > p_1 > \dots > p_N$.

Parcel. An idealized blob of air lifted adiabatically from a starting level through the environment. Its temperature along the ascent is determined by thermodynamics (conservation of entropy), *not* by the environment; the environment only enters through comparison.

Buoyancy. $b(p) = (\text{density temperature of the parcel at level } p) - (\text{density temperature of the environment at } p)$, in Kelvin. Positive b pushes the parcel up. Density temperature is a smooth algebraic function of temperature, moisture and pressure.

CAPE (convective available potential energy, J kg^{-1}): the work the buoyancy force can do on the parcel, an integral of b described below.

LNB (level of neutral buoyancy): the level where an ascending parcel’s buoyancy returns to zero — where it “detrains” (exits). The temperature there, the *outflow temperature* T_O , sets the Carnot efficiency of the storm-as-heat-engine.

SST: sea-surface temperature. **MSL**: sea-level pressure far from the storm.

2.1 The work integral and the CAPE conventions

Fix a column: levels $p_0 > \dots > p_N$ (the code retains levels with $p_j > 50$ hPa; $N + 1 = 28$ for the soundings here), environmental temperatures T_j and moisture r_j . For a parcel specified by its starting pressure, temperature and moisture, thermodynamics produces a buoyancy value b_j at every level: below its condensation level by an explicit smooth formula, above it by inverting a smooth, strictly monotone entropy relation (a guarded Newton iteration in the code).

Observation 1. *Each b_j is a continuous — indeed, away from one benign switch, real-analytic — function of the parcel parameters. (The Newton stopping tolerance of 10^{-3} K and a formula switch at the condensation level introduce micro-jumps of order $\lesssim 10^{-3}$ K in b_j ; these propagate to $\lesssim 10^{-2}$ hPa in the pressure map, $50\times$ below the solver tolerance, and are accounted for separately in §7.4.)*

Define the *running work integral* from the parcel’s starting pressure q_0 up to a trial “outflow” pressure $q \leq q_0$:

$$W(q) = R_d \int_q^{q_0} b(p) \, d \ln p, \quad (1)$$

where $R_d = 287$ in SI units and $b(p)$ is the piecewise-linear interpolant of the b_j . The code evaluates (1) by the trapezoid rule $R_d(b_j + b_{j-1})(p_{j-1} - p_j)/(p_{j-1} + p_j)$ on grid intervals, plus a partial term up to an interpolated zero of b at the top — i.e., exactly the integral of the piecewise-linear reconstruction, with the terminal point located *sub-grid* by linear interpolation. W is a C^1 -in- q , piecewise-quadratic function whose critical points are the zeros of b : local maxima at down-crossings ($b : + \rightarrow -$, candidate LNBs) and local minima at up-crossings. Figure 1 shows the generic anatomy: one positive region of b , one hump of W , an unambiguous LNB at the argmax.

Remark 1 (the p_{top} truncation can clip the LNB). *The retained column ends at the last level above $p_{\text{top}} = 50$ hPa — here 70 hPa. For the most intense columns the saturated parcel is still buoyant at 70 hPa (its true zero crossing lies above the cutoff); the code then sets INB to the top retained level and reports $\text{LNB} = 70$ hPa, $T_O = T(70 \text{ hPa})$. In the 2024 benchmark this affects 23/7093 converging columns (0.3%) — precisely the extreme tail, every one with $V_{\text{max}} > 86 \text{ m s}^{-1}$. The bias has two competing parts: the true LNB sits higher, where the*

Generic column (January, eastern tropical Pacific, 10°N 108°W): one positive region, unambiguous LNB

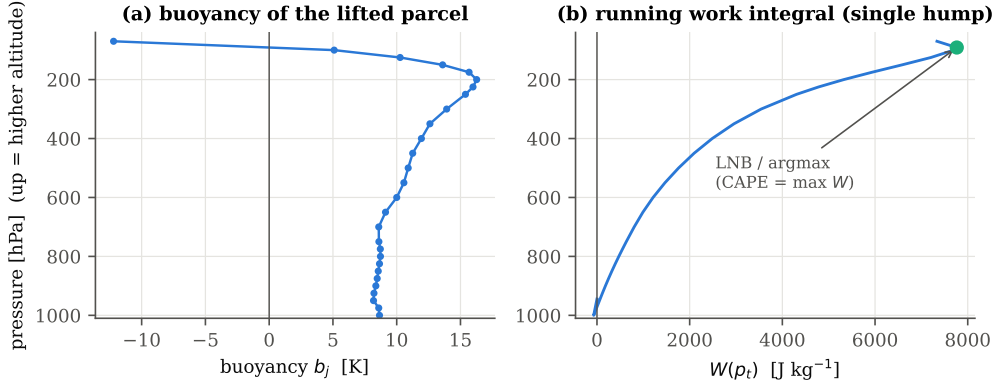


Figure 1: A generic strong-PI column (January, eastern tropical Pacific, 10°N 108°W; the saturated parcel of §2.2, $V_{\max} = 72.5 \text{ m s}^{-1}$). (a) Buoyancy of the lifted parcel at the 28 retained grid levels: positive through the troposphere, crossing zero between the 100 and 70 hPa levels. (b) The running work integral $W(q)$: a single dominant hump; CAPE (7764 J kg⁻¹) is its maximum, attained at the interpolated LNB (91.2 hPa). For such profiles every convention below coincides.

stratospheric environment is warmer, so the clipped T_O is too cold (efficiency, hence $V_{\max}$, overestimated), while the uncaptured positive area above 70 hPa underestimates CAPE. It cannot be quantified from the shipped sample, whose data itself stops at 50 hPa; assessing it requires source data with higher levels (available in ERA5, and present in the PR #77 profile) plus a p_{top} sensitivity test — bearing in mind that the reversible parcel model itself becomes questionable in the stratosphere, which is presumably why `pcmin` truncates at all, and that under the legacy selector a deeper column would add more marginal pockets to flicker (under `max-work` it is harmless). This truncation bias is orthogonal to the discontinuity studied here, but a robust API should flag a clipped LNB (INB at the profile top) rather than report it as an ordinary outflow level.

Remark 2 (the two negative tails of W). *In the W figures the curve dips negative at both ends, for different reasons. Above the LNB, $b < 0$ and W falls away from its maximum — mildly at first, then steeply in the stratosphere (to $\approx -8000 \text{ J kg}^{-1}$ by 70 hPa in the #77 sounding; the figures clip this) — which is exactly what makes the LNB the argmax. Near 1000 hPa the negativity is bookkeeping, not ascent physics: W is anchored at the parcel’s launch pressure $q_0 = \min(x, 1000)$, which for the storm-core parcels lies above the lowest data level because the trial central pressure x is below 1000 hPa. For $q > q_0$ (below the launch point) $W(q) = -R_d \int_q^{q_0} b \, d \ln p < 0$ when the parcel is buoyant at the bottom: displacing it downward costs work. The code carries this segment as a signed “partial area between the parcel pressure and the lowest level” correction (present identically in `pcmin.m`); the parcel never visits it. The anchor itself, $W(q_0) = 0$, is a candidate terminal point — which is why $E_{\max} \geq 0$ below holds with no clamp despite both tails.*

Turning W into a single number requires choosing a terminal point, and here conventions diverge. Three natural choices:

Convention 1 (value function). $E_{\max} = \max_{q \in [p_N, q_0]} W(q)$, the value of the parametric maximization; the LNB is the argmax. Since $q = q_0$ is a candidate with $W(q_0) = 0$, automatically $E_{\max} \geq 0$; no clamping is needed.

Convention 2 (ballistic / reachable). $E_{\text{reach}} = \max\{W(q) : q \in \mathcal{R}\}$, where $\mathcal{R}$ is the connected component of $\{W > 0\}$ containing q_0 . A parcel starting *at rest* converts work to kinetic energy and stalls where W first returns to zero; E_{reach} is the best energy among levels such a parcel can reach unaided.

Convention 3 (topmost-positive-level — the code’s rule). Let $\text{INB} = \max\{j : b_j > 0\}$ (a *grid* index). The code integrates the signed work up to level INB, adds the partial area up to the interpolated zero above it, and clamps the result at zero:

$$E_{\text{top}} = \max(W(z_{\text{top}}), 0),$$

where z_{top} is the topmost down-crossing of the piecewise-linear b (or the profile top if $b_N > 0$). The clamp $\max(\cdot, 0)$ turns a possibly negative signed integral into 0; it *is* reachable here, and is half of the issue-77 mechanism.

To be clear about provenance: Convention 3 is *not* a porting divergence. The selector, the clamp, and every formula above are inherited line-for-line from Emanuel’s `pc_min.m` (whose comment reads “Find maximum level of positive buoyancy, INB”), translated verbatim in the very first `pyPI` commit (April 2020) and never altered since. Neither the CAPE definition of Emanuel (1994, Eq. 6.3.6) nor Bister and Emanuel (2002) specifies a rule for profiles with multiple buoyant layers [4, 1]; the topmost-positive-node selection is an undocumented implementation choice of the reference code, which the Python port faithfully reproduces — including its discontinuity.

Pointwise, $E_{\text{top}} \leq E_{\text{max}}$ and $E_{\text{reach}} \leq E_{\text{max}}$: Convention 3 evaluates W at one particular candidate, Convention 1 at the best one. Figure 2 shows the four archetypal profiles that span all the ways the three can relate. (a) For a single-hump W — the overwhelmingly common case — all three agree exactly. They separate only when the buoyancy has a second positive layer aloft, and then in three distinct ways. (b) If the upper cell is *dominated* (its local maximum of W is lower) but no energy barrier intervenes ($W > 0$ throughout), then $E_{\text{max}} = E_{\text{reach}}$ at the lower cell while E_{top} , bound to the *topmost* crossing, returns the smaller upper value: it undercounts by charging the full traversal cost of the valley for a destination that was not worth reaching. (c) If the upper feature is marginal *and* sits behind a barrier ($W < 0$ in between) — the issue-77 configuration — the signed integral to the top is negative and the clamp returns $E_{\text{top}} = 0$, while $E_{\text{max}} = E_{\text{reach}}$ sit unperturbed at the lower cell; worse, as §3 shows, E_{top} here flips discontinuously between the two regimes as the marginal feature’s sign flickers. (d) If the upper cell is *dominant* but behind a barrier, $E_{\text{max}} = E_{\text{top}}$ at the top while the ballistic E_{reach} stalls at the lower cell: the one configuration where the reachability convention parts company with the other two.

Which convention does the physics of PI want? The word “CAPE” is misleading here. In an ordinary sounding it is read as kinetic energy available to a freely rising parcel, for which the ballistic Convention 2 is the natural semantics. The PI quantity is instead *hurricane CAPE*: in the Bister–Emanuel derivation the CAPE difference is an alternative expression for the thermodynamic work around the idealized storm cycle [1, 2], and eyewall ascent in a mature vortex is driven primarily by frictional convergence — organized, forced, predominantly hydrostatic — not by a parcel coasting on its own buoyancy [3]. An intervening negatively buoyant layer must still be *charged against the work* (it reduces W), but it is not a hard reachability barrier for the forced secondary circulation. That is precisely Convention 1: every layer costs what it costs, and the outflow level is the one maximizing extractable work. This distinction drives the recommendation in §7.

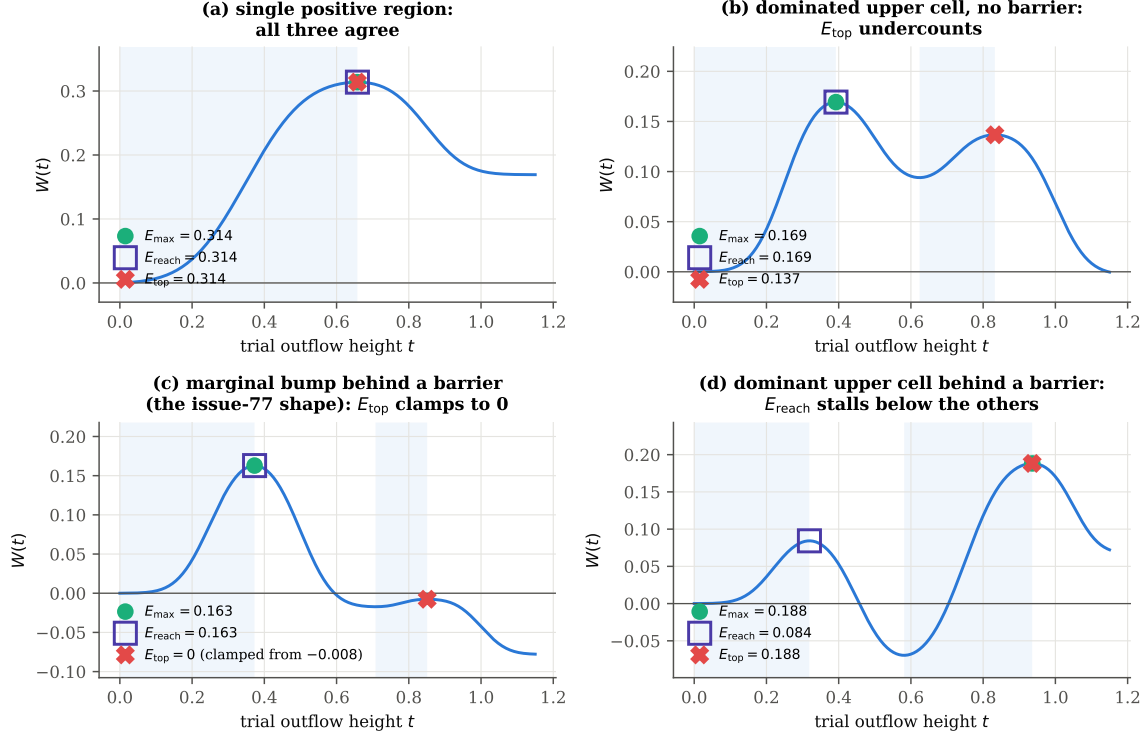


Figure 2: The three CAPE conventions on four archetypal smooth profiles (shaded bands: layers of positive buoyancy; curves: the running work integral W). Markers show each convention’s terminal point and value: $E_{\max}$ (dot), E_{reach} (open square), E_{top} (cross). (a) One positive region: all agree. (b) Dominated upper cell, no barrier: $E_{\text{top}} < E_{\max} = E_{\text{reach}}$. (c) Marginal bump behind a barrier (the issue-77 shape): E_{top} is clamped to 0 from a negative signed integral; $E_{\max} = E_{\text{reach}}$ ignore the bump. (d) Dominant upper cell behind a barrier: $E_{\text{reach}} < E_{\max} = E_{\text{top}}$.

2.2 The fixed-point map

The solver seeks the storm’s central pressure x (hPa) at the radius of maximum winds. Given a trial x , it computes three CAPE values with the environment held fixed: E_A (an x -independent ambient surface parcel), $E_M(x)$ (the ambient surface parcel displaced to pressure x , which enriches its moisture — the “eyewall parcel”), and $E_{MS}(x)$ (a parcel saturated at the SST at pressure x — the storm-core parcel, whose LNB also furnishes the outflow temperature $T_O(x)$). These combine into

$$g(x) = \text{MSL} \cdot \exp\left(-\text{CAT}(x) / (R_d \bar{T}_v(x))\right), \quad (2)$$

$$\text{CAT}(x) = \max\left((E_M - E_A) + \frac{1}{2} \kappa \frac{\text{SST}}{T_O} (E_{MS} - E_M), 0\right),$$

with $\kappa = C_k/C_D = 0.9$ and $\bar{T}_v$ a smooth positive function of x . The released code runs Picard iteration $x_{n+1} = g(x_n)$ from $x_0 = 970$, stopping when consecutive iterates satisfy $|x_{n+1} - x_n| \leq 0.5$ hPa, capped at 200 steps. (A two-back comparison $|x_{n+1} - x_{n-1}|$ appears only in the recently added cycle-detection patch, not in the stopping rule.) All CAPE evaluations use Convention 3. Note that even for continuous g , Picard iteration requires local contractivity $|g'(x_*)| < 1$; existence of a fixed point alone is not enough — and for discontinuous g , neither existence nor convergence is available.

Away from the pathologies below, g is a mild contraction: on the failing sounding its slope off the jump is 0.34–0.46, and generic columns converge geometrically (Fig. 3).

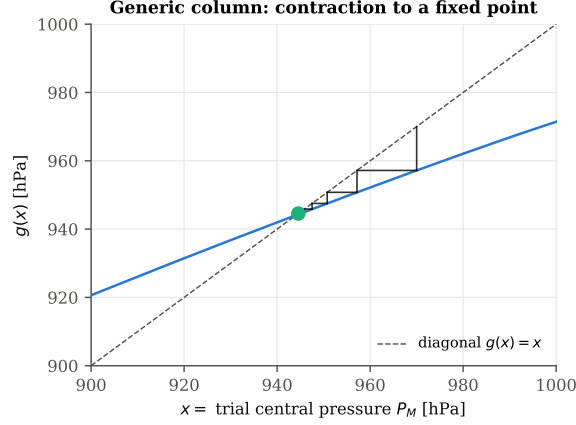


Figure 3: The map g for the generic column of Fig. 1, with the cobweb of the iteration from $x_0 = 970$: contraction to a fixed point at 944.6 hPa in seven steps.

3 A smooth toy model: what is genuinely discontinuous

Everything essential can be exhibited with C^∞ data, which settles at the outset that grid resolution, quadrature order, and interpolation smoothness are *not* the cause (a claim tested directly on the real sounding in §6.1). Take a smooth family of buoyancy profiles on a height coordinate t ,

$$b(t) = \underbrace{0.6 e^{-\left(\frac{t-0.22}{0.10}\right)^2}}_{\text{main convective cell}} - \underbrace{\delta e^{-\left(\frac{t-0.5}{0.12}\right)^2}}_{\text{stable "valley"}} + \underbrace{\varepsilon e^{-\left(\frac{t-0.8}{0.06}\right)^2}}_{\text{upper feature}}, \quad (3)$$

with $W(t) = \int_0^t b$, and evaluate the three conventions along two sweeps (Fig. 4). The sweeps animate the archetypes of Fig. 2: sweep 1 carries the profile into and out of archetype (c) as the marginal bump’s sign flickers; sweep 2 deforms archetype (b) into archetype (d) as the valley deepens and the barrier closes.

The toy isolates three distinct behaviors:

1. $E_{\max}$ **is continuous, always** — Berge’s maximum theorem in its simplest form: $W(\cdot; \lambda)$ is a jointly continuous family over a fixed compact interval, so $\lambda \mapsto \max_q W(q; \lambda)$ is continuous *even where the argmax jumps*. The argmax correspondence is only upper hemicontinuous: at a dominance exchange the maximizer teleports (by 455 hPa in the real-data example of §6), but the value passes through with a kink, not a jump.
2. E_{reach} **can jump for real physical reasons**, but only at a codimension-one tangency: the local minimum of W in the valley touches zero, the set $\{W > 0\}$ changes topology, and the reachable energy jumps. A bona-fide first-order transition — but of the *free-convection* problem, which is not the PI problem (§6).
3. E_{top} **jumps at events with no physical content**: the sign of b at its topmost zero is toggled by an $O(\varepsilon)$ feature carrying $O(\varepsilon)$ energy, dominated by orders of magnitude, and (in sweep 1) sitting behind a closed barrier. The discontinuity locus of the “topmost positive point” selector strictly contains the physical loci of items 1–2 and is enormously larger. The same holds for smooth fields in general: an upper positive pocket is born at a fold ($b = 0$, $\partial_t b = 0$ simultaneously), and the uppermost-root diagnostic jumps there even though b is C^∞ and the value function is not affected unless the new pocket actually dominates.

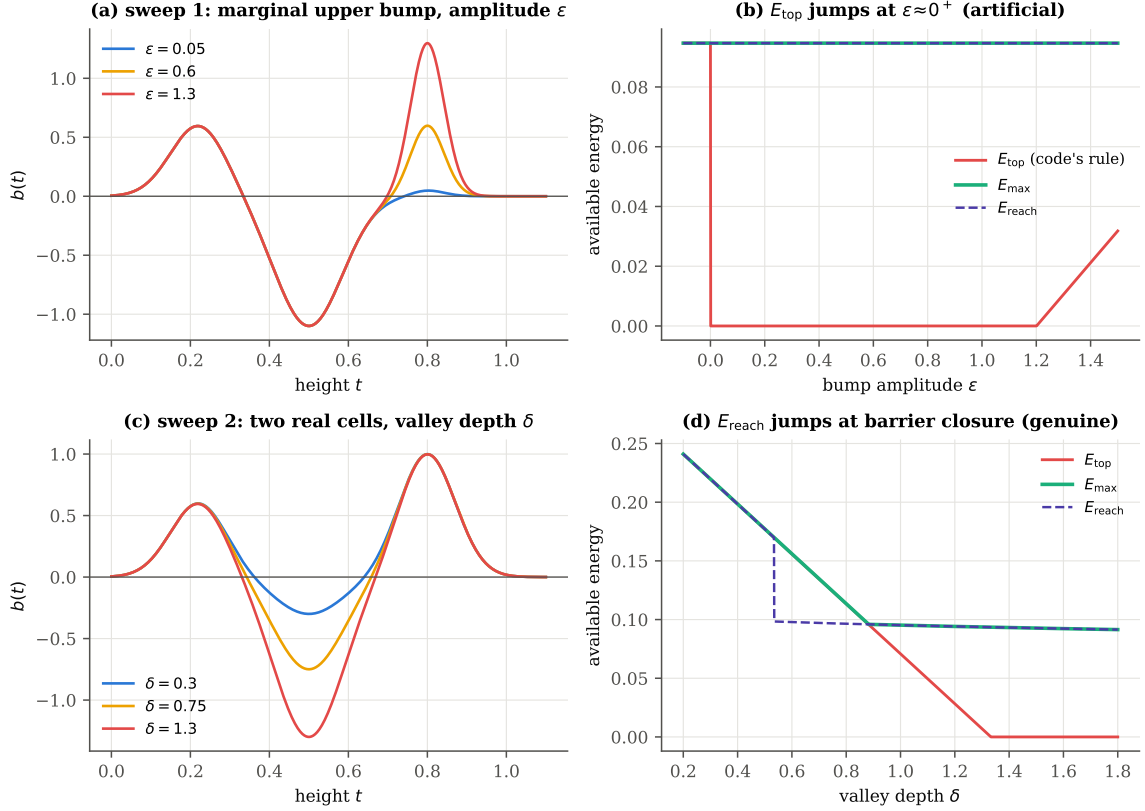


Figure 4: The smooth toy model (3). **Sweep 1** (top row; $\delta = 1.1$ fixed): the amplitude ϵ of a *marginal* upper bump crosses zero. E_{top} jumps from 0.095 to 0 at $\epsilon = 0^+$: the instant the bump’s peak becomes positive, the topmost zero of b teleports above it and the terminal value $W \approx -0.13 < 0$ is clamped to 0. E_{max} and E_{reach} do not react at all — the bump is dominated *and* unreachable. This is the issue-77 mechanism in pure form. **Sweep 2** (bottom row; two genuinely competing cells, valley depth δ varies): E_{max} is continuous with a kink where the argmax exchanges humps ($\delta = 0.881$), while E_{reach} has a *genuine* downward jump at $\delta = 0.534$, where the energy barrier closes and the upper cell becomes ballistically unreachable.

3.1 Why the symptom is a bit-exact 2-cycle

Insert a downward jump into an otherwise flat contraction so that the gap straddles the diagonal: $g(x) - x > 0$ on the left branch, < 0 on the right. No fixed point exists, every orbit must cross the jump forever, and because g is nearly constant on each side (slope $\ll 1$), the branch values $g(A) \approx B$ and $g(B) \approx A$ form a *superattracting* 2-cycle: $g \circ g$ contracts strongly onto each branch point, so the alternation converges bitwise (Fig. 5). The observed 16-digit lock is the canonical dynamics of a gap-straddling contraction — an attracting border-collision cycle, not a curiosity of the sounding.

4 The failure in the wild: Hurricane Francine, issue #77

4.1 Provenance

The failing sounding is the environment of an *actual tropical cyclone*. PR #77 records 28.6°N, 92.1°W at 18 UTC on September 11, 2024 — and the National Hurricane Center best track places Hurricane Francine at exactly that position and time, at 80 kt a few hours before its Louisiana landfall [7, 6]. The thermodynamic column is a *counterfactual-climate sensitivity*

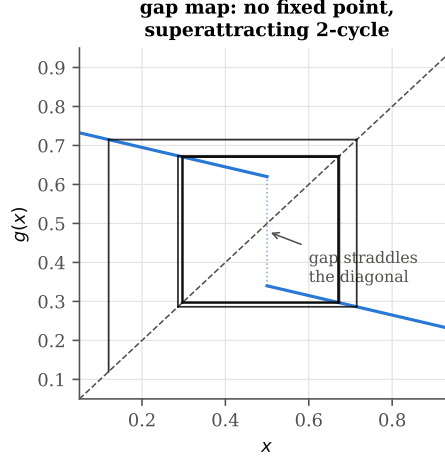


Figure 5: Abstract gap map: piecewise-linear contraction (slope 0.25) with a downward jump straddling the diagonal. Orbits fall into the unique 2-cycle $\{g(B), g(A)\}$; the cycle is superattracting because $|g'(A)g'(B)| \ll 1$.

member associated with the storm (SST perturbed by -1.6 K, mapped to a 1994/day-263 reference), not an in-situ measurement — a distinction that matters physically but not to the numerical diagnosis. The failure thus arises while asking the operational question this code exists to answer: *how strong could this storm have been in a cooler climate?* For that member the solver returns “missing”. Column parameters: SST = 28.203°C , MSL = 1014.97 hPa, 37 levels (28 retained).

4.2 The map has no fixed point

Scanning g at 0.002 hPa resolution over $[946, 958]$ (Fig. 6) yields:

Observation 2. *On $[946, 958]$ the map g has exactly one discontinuity, a downward jump of 0.837 hPa at $x_d \simeq 950.8289947515$ hPa. The residual $F(x) = g(x) - x$ satisfies $F > 0$ for $x < x_d$ and $F < 0$ for $x > x_d$ (Fig. 8b): the only sign change occurs across the jump, so g has no fixed point. Away from x_d , g is smooth with slope 0.34 – 0.46 . The iterates converge to the 2-cycle $\{A, B\}$ straddling x_d .*

4.3 One-sided anatomy of the jump

Evaluating all internal quantities 10^{-9} hPa on either side of x_d isolates the discontinuous term:

quantity	$x_d - 10^{-9}$	$x_d + 10^{-9}$
buoyancy sample at 350 hPa (K)	$+7.58 \times 10^{-11}$	-7.99×10^{-11}
highest positive index	18 (350 hPa)	10 (750 hPa)
diagnosed LNB (hPa)	350.0000	741.4664
eyewall CAPE E_M (J kg^{-1})	0	224.23146
saturated CAPE E_{MS} (J kg^{-1})	8565.00966	8565.00966
saturated outflow T_O (K)	206.1096884	206.1096884
$g(x)$ (hPa)	951.3363394	950.4982353
$F(x) = g(x) - x$ (hPa)	+0.5073447	-0.3307594

The saturated terms — and hence the outflow temperature entering (2) — are smooth; the discontinuity is entirely in the eyewall parcel’s E_M , through the selector of Convention 3. (The

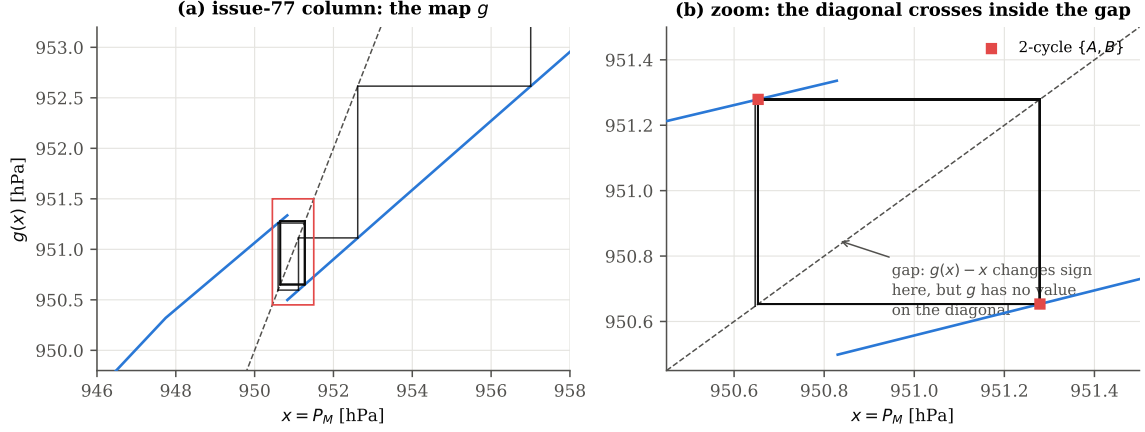


Figure 6: The issue-77 map (Hurricane Francine’s environment, -1.6 K counterfactual). (a) g on $[946, 958]$ with the iteration cobweb from $x_0 = 970$; the red box marks the zoom. (b) Zoom: two smooth branches separated by a 0.837 hPa gap; the diagonal passes through the gap; the iteration locks into the bit-exact 2-cycle (red squares).

10^{-11} K values demonstrate the one-sided limit; they are not the amplitude of the atmospheric feature. At the two cycle phases the 350 hPa buoyancy is $+0.0137$ and -0.0350 K: marginal and energetically negligible, but not floating-point noise.)

Figure 7 shows the geometry. The eyewall parcel’s buoyancy has a healthy positive region below 750 hPa (the main hump of W , worth ≈ 222 – 224 J kg^{-1} depending on the trial pressure), a deep negative valley from 750 to 400 hPa (-287 J kg^{-1}), and the marginal excursion at the 350 hPa node. When that node’s sign flickers positive, the selector teleports its terminal point from ≈ 745 to 350 hPa, the signed integral becomes $226.6 + 0.01 - 287.5 = -60.9$ J kg^{-1} , and the final clamp $\max(\cdot, 0)$ maps it to 0. Composing the three steps,

$$b \mapsto \text{INB} = \max\{j : b_j > 0\} \mapsto W(z_{\text{top}}) \mapsto \max(\cdot, 0),$$

the first arrow is the discontinuous one; the clamp then pins one side to the constant 0, maximizing the jump. Note also (Fig. 7b) that $W(350 \text{ hPa}) = -61 \text{ J kg}^{-1} < 0$: even a ballistic parcel could never reach the level whose sign decides between 224 and 0 — it would stall inside the valley. Under *every* available physical semantics ($E_{\text{max}}, E_{\text{reach}}$) the answer on both sides of the flicker is the same $\approx 222 \text{ J kg}^{-1}$.

4.4 The second instance, and the control

The 1995 sounding of the repository’s regression tests fails by the identical geometry (Fig. 9): its 400 hPa sample flickers, E_M flips $0 \leftrightarrow 203.05 \text{ J kg}^{-1}$, the residual jumps $+0.615 \rightarrow -0.225$ hPa, and the iteration cycles at $961.017 \leftrightarrow 961.671$ hPa. An adjacent-day control profile has an ordinary continuous crossing and converges — the failure is a property of marginal columns, not of the data source. Both failing columns are *marginal* in the precise sense that the ambient CAPE E_A is tiny (1.19 J kg^{-1} for #77; exactly 0 for both 1995 profiles under the released quadrature): selector topology matters most near such boundaries, where the clamp is live and humps are shallow.

4.5 Why the upstream “rescue” is not a solution

A recent upstream patch detects the 2-cycle and collapses it to its midpoint (≈ 950.966 hPa here), exiting through the normal convergence path. But the one-sided residuals at the

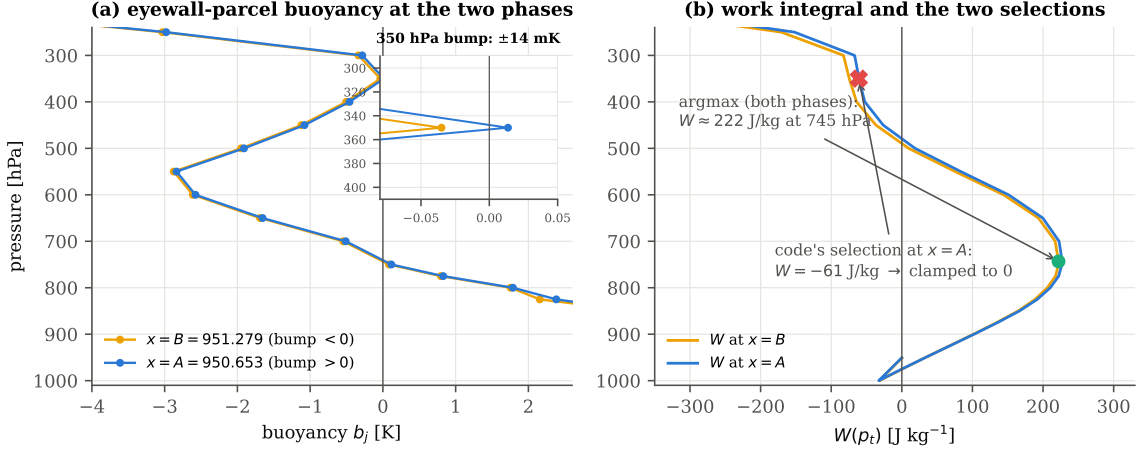


Figure 7: The issue-77 mechanism. (a) Eyewall-parcel buoyancy at the two cycle phases (curves nearly coincide); inset: the marginal excursion at 350 hPa whose sign differs between phases by ± 14 mK. (b) The corresponding work integrals: the argmax (222 J kg^{-1} at 745 hPa, green) is identical in both phases; the code’s terminal point at phase *A* (red cross) sits at 350 hPa where $W = -61 \text{ J kg}^{-1}$, clamped to 0.

jump are $+0.507$ and -0.331 hPa: *there is no root there to approximate*. The midpoint is a convention for returning a number, not a solution of $g(x) = x$; moreover the thermodynamic fields ultimately reported are evaluated at one cycle phase, an arbitrary parity-dependent choice worth $\sim 1.2 \text{ m s}^{-1}$ in V_{max} . Cycle detection remains useful as telemetry and as a guard against burning 200 iterations, but it treats the symptom. The disease is the discontinuity.

5 What is generic, what is unusual

Two separate questions: how common are jumps of g , and how common is the *fatal configuration* (diagonal crossing inside a jump)?

We scanned g at 0.1 hPa resolution over $x \in [900, 1005]$ for 1,200 randomly sampled converging columns of the repository’s 2024 North-Atlantic sample (Fig. 10a). **Jumps are generic:** 239/1200 columns (20%) carry at least one jump, with sizes up to $\sim 1 \text{ hPa}$ — unsurprising, since the tropical upper troposphere hovers near moist neutrality ($b \approx 0$ over deep layers), so marginal sign flickers under the x -dependent parcel moisture are everywhere. **The fatal configuration is rare:** in zero of the 1,200 columns does the diagonal crossing fall inside a jump, consistent with the full 7,093-column sample containing no convergence failures.

How close is “close”? Perturbing the Francine column’s SST further (Fig. 10b), the no-fixed-point window is $\sim 0.04 \text{ K}$ wide out of 6 K scanned. This explains where the failure surfaced: an SST-attribution sweep around a real storm samples SST densely and will eventually land inside some column’s window. Single climatological runs rarely hit it; parameter sweeps over live cases — where a silently missing value is most costly — will. The alignment is also less improbable than it looks: the iteration actively ferries orbits from both sides back across the jump, so once the crossing is inside the gap the cycle is *attracting*, and a jump elsewhere passes unnoticed (either it does not straddle zero, or the induced cycle amplitude is below the 0.5 hPa tolerance and is silently accepted as convergence).

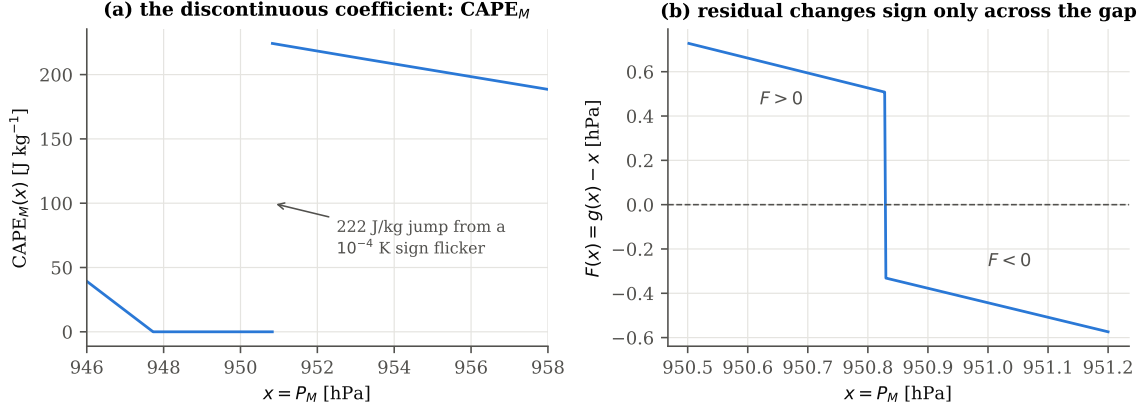


Figure 8: (a) E_M as a function of the trial pressure x : piecewise smooth with the cliff at x_d (the structure at $x < 948$ is a second, harmless flicker of the same kind). (b) The residual $F = g - x$ near x_d : the only sign change is across the gap; no root exists.

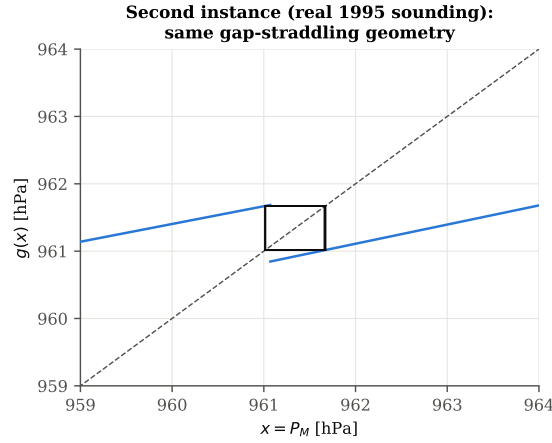


Figure 9: The second known failing sounding (real 1995 profile): the same gap-straddling geometry and 2-cycle.

6 To what extent is the discontinuity physical?

Argmax-type constructions *do* have genuine discontinuities, and one must not repair those away. The precise accounting:

The maximizer may genuinely jump; the value may not. CAPE-as-energy is the *value* of $\max_q W(q; \lambda)$; the LNB and outflow temperature are the *maximizer*. By the maximum theorem the value is continuous in the data; the maximizer is only upper hemicontinuous and jumps at dominance exchanges. Figure 11 shows this on a real column: a 0.4% moisture change teleports the argmax $870 \rightarrow 410 \rightarrow 180$ hPa while the value crosses each exchange with a kink. Physically the maximizer's jump is a real bifurcation — the preferred detrainment layer switches — and any code reporting T_O must live with it. (In (2), T_O comes from the *saturated* parcel, whose W is single-humped with a $\sim 40\times$ dominance margin for storm-relevant SSTs, so this genuine discontinuity is dormant in practice; see §7.4.) The maximizing level is in any case metadata rather than a prediction: where real eyewall air detrains depends on dynamics and mixing absent from a reversible parcel calculation.

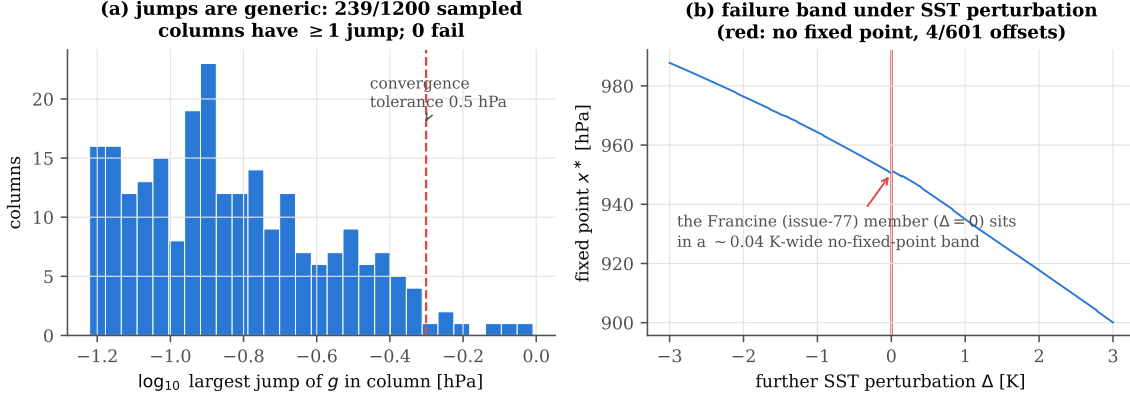


Figure 10: (a) Largest jump of g per column over 1,200 sampled real columns (only the 239 with a detected jump shown). Jumps are common; none coincides with its column’s diagonal crossing. (b) The Francine column under further SST perturbation: the fixed point moves smoothly except in a ~ 0.04 K band (red) where it does not exist; the issue-77 member sits inside it.

Reachable energy can genuinely jump — in a different problem. Under ballistic semantics (Convention 2) the value itself jumps when an energy barrier closes at a tangency (toy sweep 2). That is legitimate physics — of *spontaneous free convection*: an isolated parcel launched from rest does not teleport through a barrier. It is not the physics of hurricane PI, whose eyewall ascent is forced by the storm’s own frictionally convergent circulation [3, 1]. A reachability-restricted CAPE answers a different question and should at most be a separately named diagnostic, never the PI default (§7).

The code’s discontinuity is neither of these. Convention 3 jumps whenever the topmost sign of b flickers — a strict superset of the physical loci above. At the issue-77 event specifically: the deciding feature carries $O(10^{-2}) \text{ J kg}^{-1}$ against a 224 J kg^{-1} decision (no energetic semantics); the selected terminal level has $W = -61 \text{ J kg}^{-1}$, unreachable even ballistically (no reachability semantics); and both $E_{\max}$ and E_{reach} are constant through the event. If the “topmost level” rule encodes an intention — “outflow happens at the highest level convection can reach” — then $E_{\max}$ ’s argmax honors that intention on every profile where the top level is attainable and worth attaining; where the rules differ, E_{top} is charging the parcel an entry fee for a room it never enters.

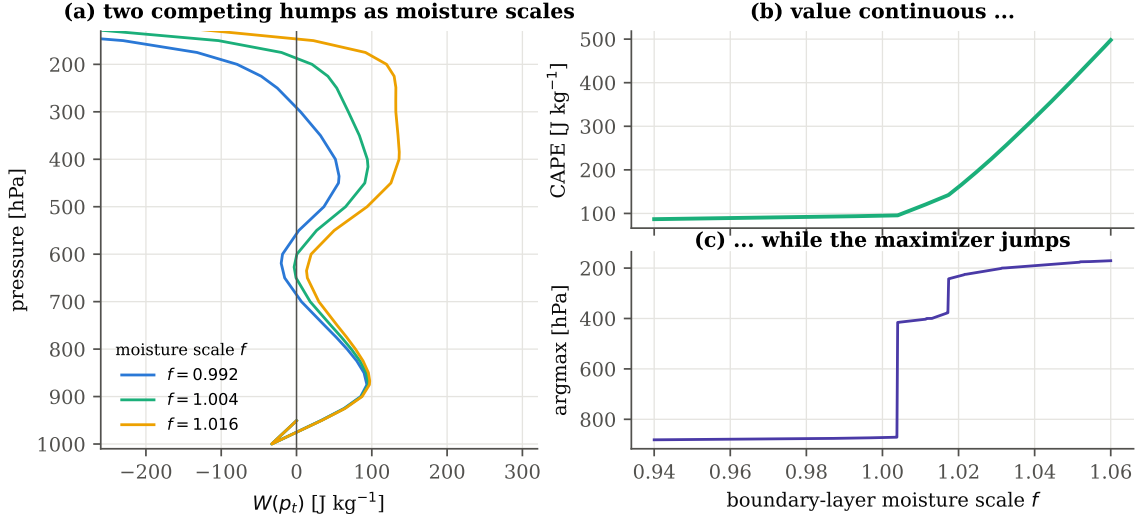


Figure 11: Berge’s theorem on a real column (June, 28.5°C SST). (a) The eyewall-parcel work integral has two competing humps 450 hPa apart. (b,c) Sweeping boundary-layer moisture by $\pm 6\%$: $\text{CAPE} = \max_q W$ is continuous through both dominance exchanges (kinks), while the argmax jumps. Value and maximizer have different continuity classes; only the maximizer’s jumps are physical.

feature	status	reason
multiple buoyant layers	physical possibility	stratification can create several neutral levels
birth of a remote positive pocket	genuine mathematics	a smooth field can undergo a fold/tangency
maximizing LNB jumping	genuine (variational)	work maxima exchange dominance; the common value stays continuous (Berge)
reachable energy jumping	genuine, wrong problem	barrier closure is free-convection physics, not forced-ascent PI
uppermost-positive-node jumping	genuine mathematics, wrong value rule	that node need not maximize work nor be reachable
E_M jumping $0 \leftrightarrow 224 \text{ J kg}^{-1}$	artifact	selector + signed integration + clamp amplify a one-bit sign
pressure two-cycle	solver artifact	a discontinuous map is iterated as though contractive

6.1 What the cause is not

Three natural alternative hypotheses were tested and refuted, and deserve one paragraph so nobody re-litigates them. *Quadrature order*: the trapezoid rule with sub-grid terminal crossings integrates the piecewise-linear reconstruction exactly and handles simple crossings continuously; the jump survives with the same quadrature under Convention 1, and disappears with the same quadrature under a changed *selection* — the quadrature is innocent. *Grid resolution / interpolation smoothness*: rebuilding the Francine environment with a C^1 monotone-cubic interpolant of T, r in $\ln p$ resampled at 5 hPa resolves the 350 hPa feature (it is real, and $4\times$ taller than the coarse grid shows) and happens to converge — but the map still has an 0.83 hPa jump, merely relocated to 951.4 hPa, off the diagonal *by luck*; the same resampling with linear interpolation leaves the jump straddling the diagonal and still fails. No reconstruction, of any smoothness class, can remove the discontinuity of E_{top} : §3 exhibits it for C^∞ data.

Floating point: the one-sided 10^{-11} K values are ordinary continuity of $b_{18}(x)$ through zero, not roundoff; the feature itself is tens of millikelvin at the cycle phases. Discretization's real role is secondary: it quantizes *where* flickers can fire (at nodes) and lends a ~ 100 hPa-wide lever arm to a narrow feature, but the selection rule decides *that* they fire.

7 The repair

7.1 Maximum-work CAPE: one changed selection

Replace Convention 3 by Convention 1 in all three CAPE evaluations of (2): accumulate the *same* running signed work W over the *same* piecewise-linear reconstruction with the *same* sub-grid crossing candidates, and take the maximum over candidate terminal points instead of the value at the topmost crossing. Return the argmax as the LNB (with an explicit tie rule). No clamp is needed. Nothing is skipped and nothing is free: every negative-buoyancy layer reduces W along the way. The change is one selection, not new machinery — and for single-hump profiles it reproduces the legacy value exactly.

This is also the physically correct convention for PI (§6): hurricane CAPE is thermodynamic work around a forced cycle, so the outflow level is the one maximizing extractable work, with traversal costs charged but no ballistic reachability requirement. The ballistic (first-connected-lobe) rule was also implemented as a proof-of-mechanism: it too removes the failures, but it answers the free-convection question — and it perturbs the healthy control column by 1.2 hPa where max-work moves it by 0.0035 hPa. Maximum signed work is the minimal-change, correct-semantics repair.

7.2 Existence, and solving for a root

Proposition 1. *With $E_{\max}$ in place of E_{top} in all three CAPE evaluations, and modulo the micro-jumps of §7.4, the map g of (2) is continuous on $[400, \text{MSL}]$ wherever the saturated parcel's argmax is unique, and $g \leq \text{MSL}$ so that $F(x) = g(x) - x$ satisfies $F(\text{MSL}) \leq 0$. If in addition the column satisfies the non-hypercane endpoint condition $g(400) > 400$, then $F(400) > 0$ and the intermediate value theorem gives a fixed point in $(400, \text{MSL}]$.*

Proof sketch. Each b_j is continuous in x (Observation 1); W is jointly continuous on a compact set; $E_{\max} = \max_q W$ is continuous (maximum theorem); E_A is constant and $\bar{T}_v$ smooth and positive; $T_O(x)$ is continuous wherever the saturated argmax is unique; $\max(\cdot, 0)$ and $\exp$ preserve continuity. $\text{CAT} \geq 0$ gives $g \leq \text{MSL}$. $\square$

Both hypotheses are explicit for a reason. If $g(400) \leq 400$ the physical floor has been reached and a distinct domain flag should be returned, not a root. And because residual non-smoothness survives at the 10^{-2} hPa scale (§7.4), a bracketed solver (bisection or safeguarded Brent on F) should declare success only after an explicit residual check $|F(x_*)| \leq \varepsilon_F$ — bisection on a residual with an undetected jump converges to the jump, exactly the trap of Observation 2. With those guards, bracketing terminates deterministically in ~ 40 evaluations, needs no contractivity assumption, and cannot enter a period-two orbit. (The legacy Picard loop also converges here, since $|g'| \leq 0.47$ off the former jump — but bracketing turns an empirical slope into a guarantee.)

7.3 Results

- **Failing cases (exact roots).** #77/Francine: $x_* = 950.3136889226$ hPa, $V_{\max} = 83.9065$ m s $^{-1}$, $P_{\min} = 920.7517$ hPa, $T_O = 206.088$ K. 1995 marginal: $x_* = 960.7687$ hPa, $V_{\max} = 74.4149$ m s $^{-1}$. The legacy Picard loop with max-work CAPE also satisfies

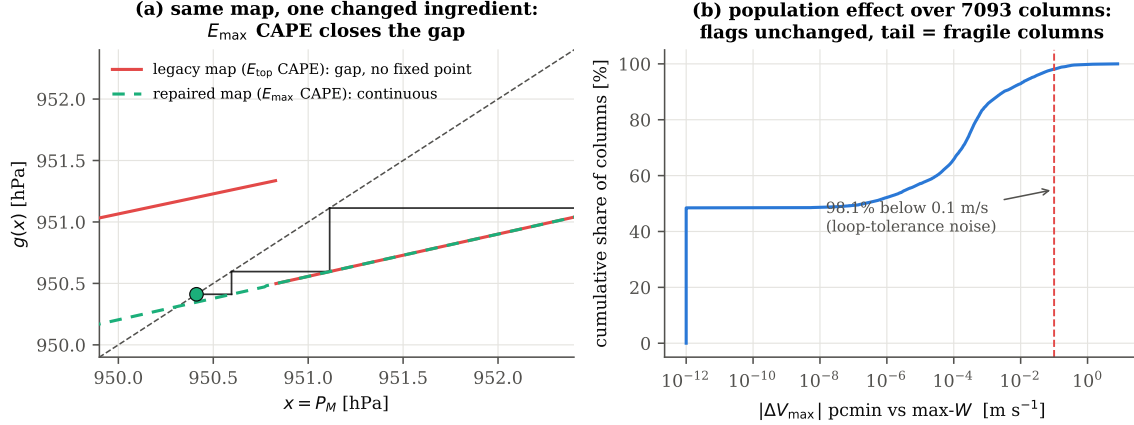


Figure 12: The repair: the same map (2) evaluated with the two CAPE conventions on the Francine column. (a) With the legacy E_{top} (red, solid) the map has the gap and no fixed point. With E_{max} (teal, dashed) it is continuous, crosses the diagonal, and the iteration converges (dot). On the right branch the two coincide to line width — red shows through the dash gaps — because there the topmost-positive level already selects the dominant lower cell; on the left branch they differ by the full effect of the 222 J kg^{-1} clamp. (b) Cumulative distribution of $|\Delta V_{\text{max}}|$ between conventions over the benchmark: 39% bit-identical (left edge, clipped to 10^{-12}), 98.1% below 0.1 m s^{-1} , and a 1.9% tail of discontinuity-adjacent columns.

its operational 0.5 hPa rule after five updates; note its fifth iterate (950.4115, residual -0.064 hPa) is the operational answer, not the root — the values above come from bracketing. For the 1995 case the corrected answer sits 1.1 m s^{-1} from what the upstream rescue reports (75.47, the physics of an arbitrary cycle phase).

- **Benchmark population** (patched compiled kernel, rescue deleted; 7,093 converging columns of the 2024 sample): **zero** flag changes; 39% of V_{max} values bit-identical; 98.1% within 0.1 m s^{-1} (the sub- 10^{-3} bulk is float summation-order noise laundered through the 0.5 hPa stopping rule, Fig. 12b); a 1.9% tail (138 columns) moves by up to 7.9 m s^{-1} . Fixed-pressure dissection shows every tail column is a dominated-hump or near-surface-sliver configuration — precisely the profiles on which the legacy convention was fragile by construction (a millikelvin perturbation could move its answer by hundreds of J kg^{-1}).
- **Reference parity and process.** Agreement with the MATLAB pmin reference is untouched on the 98% and diverges only on the fragile tail — necessarily, since the discontinuity is *in* the reference convention. Maximum-work CAPE is an intentional, argued scientific convention change: it should be reviewed upstream, versioned, and (if compatibility demands) initially exposed behind an explicit `cape_definition` option. Bit-identity on every column is not an appropriate acceptance criterion for an intentional correction; identity of flags, boundedness of the audited tail, and the residual check are.

7.4 Residual non-smoothness

What remains after the repair, all quantified and benign:

1. *Newton stopping rule* in the entropy inversion (tolerance 10^{-3} K): the returned temperature is a lagged iterate, so b_j has micro-jumps up to $O(10^{-3} \text{ K})$; measured step-to-step roughness of the repaired map is $\leq 7 \times 10^{-3} \text{ hPa}$ — $50\times$ below the solver tolerance, and

covered by the residual check. A literal continuity theorem applies to the converged thermodynamic relation, not to the last bit of a stopped Newton iterate.

2. *Condensation-level formula switch* at grid nodes: same order.
3. *Saturated-parcel argmax exchange* (the genuine Berge jump of T_O): dormant for storm-relevant profiles (dominance margin $\sim 40\times$), but a robust API should detect ties/near-ties in the saturated argmax and return a branch-ambiguity flag rather than silently jumping; if it ever fires, it is physics, and residual-checked bracketing still terminates.

7.5 Implementation and validation plan

Staged so numerical and scientific changes are separable:

- A — diagnostics, no changed outputs.** Expose a scalar evaluator for $F(x)$ and CAPE diagnostics (all buoyancy roots, selected root, cumulative work at each candidate, one-sided values); add automatic jump and period- k scans.
- B — max-work kernel.** Implement $E_{\max}$ in a transparent scalar reference, then in the compiled kernel without arithmetic reassociation; return value plus argmax metadata; classify every material difference from legacy on the full sample before choosing the default/compatibility policy.
- C — outflow semantics.** Specify the tie rule and the relation between the maximizing level and PI’s outflow temperature; detect saturated argmax exchanges; flag LNBs clipped by the p_{top} truncation (Remark 1). If ballistic reachability is wanted for other applications, expose it as a distinctly named free-convection diagnostic, never as the PI default.
- D — bracketed solver.** Brent/bisection with cached CAPE evaluations, endpoint-sign check first, distinct flags for invalid input, entropy-solver failure, no-bracket/physical floor, and branch ambiguity; success requires $|F(x_*)| \leq \varepsilon_F$.

Required regression families: the exact #77 column with one-sided values pinned; the 1995 marginal and control pair; perturbations of SST, surface humidity, MSL and each temperature level around both marginal cases; analytic one-lobe/two-lobe/tangency profiles with known roots; and the full sample with output deltas *and* LNB-branch-identity changes reported. On every successful column, verify $|F(x_*)| \leq \varepsilon_F$, that the returned CAPE equals the maximum candidate work, and that the LNB obeys the declared tie convention.

8 Final interpretation

The causal chain, in full:

marginal remote buoyancy pocket $\longrightarrow$ hard highest-positive-node selector
 $\longrightarrow$ finite CAPE jump (helped by the clamp)
 $\longrightarrow$ pressure-map jump straddling the diagonal
 $\longrightarrow$ attracting, bit-exact 2-cycle
 $\longrightarrow$ “missing” PI for a real hurricane’s counterfactual.

What is physical: multiple neutral levels, dominance exchanges between work maxima (maximizer jumps, value continuous), and barrier transitions in the free-convection problem. What is conventional: calling the uppermost root “the” LNB. What is artificial: letting a marginal, energetically negligible, dynamically unreachable, effectively one-node pocket erase $\sim 224 \text{ J kg}^{-1}$ of

lower-tropospheric work and thereby delete the fixed point of the pressure equation. The repair — maximum signed work as the CAPE value, argmax as reported metadata, residual-checked bracketing as the solver — is one changed selection and one changed loop, restores an existence proof under explicit hypotheses, and makes the two-cycle chase disappear for the right reason.

A Key numbers for the failing sounding

quantity	value
provenance	Francine env., 18 UTC 2024-09-11, 28.6°N 92.1°W, $\Delta\text{SST} = -1.6$ K
SST, MSL	28.20263672°C, 1014.96545 hPa
levels (retained > 50 hPa)	37 (28)
cycle A, B	950.6533454438, 951.2790079839 hPa ($B - A = 0.6257$)
jump location, size	$x_d \simeq 950.8289947515$ hPa, 0.837 hPa downward
one-sided residuals at x_d	+0.5073447 / -0.3307594 hPa
slope of g off the jump	0.34–0.46 (on [946, 958])
b at 350 hPa, cycle phases	+0.0137 / -0.0350 K
b at 350 hPa, at $x_d^\mp$	+7.6 $\times 10^{-11}$ / -8.0 $\times 10^{-11}$ K
E_M across x_d	0 $\leftrightarrow$ 224.23146 J kg $^{-1}$ (raw -60.9 before clamp)
E_{MS}, T_O at x_d (smooth)	8565.00966 J kg $^{-1}$, 206.1096884 K
ambient CAPE E_A	1.19 J kg $^{-1}$ (#77); 0 (both 1995 profiles)
corrected root (max-work, bracketed)	$x_* = 950.3136889226$ hPa
corrected outputs	$V_{\max} = 83.9065$ m s $^{-1}$, $P_{\min} = 920.7517$ hPa
1995 marginal: cycle	961.017 $\leftrightarrow$ 961.671 hPa
1995 marginal: corrected root	$x_* = 960.7687$ hPa, $V_{\max} = 74.4149$ m s $^{-1}$
1995 control (max-work vs legacy)	fixed point moves 0.0035 hPa (ballistic rule: 1.2 hPa)

B Reproduction

All computations were run against branch `upstream/modernize_eg` (commit `abc4b4e`) of the `tcpyPI` repository, using the committed 2024 ERA5 North-Atlantic sample and its committed outputs. The analysis scripts live in `discontinuity_analysis/scripts/`: instrumented map evaluation (`gmap.py`), fine scans (`scan.py`), the max-work prototype and full-sample validation (`maxw.py`, `validate.py`, `compare.py`, `dissect.py`), the survey and SST sweep (`datagen.py`), the dominance-exchange hunt (`humphunt.py`), the reconstruction refutation (`refine.py`), and figure generators (`plots.py`, `plots2.py`). Every headline quantity (cycle values, jump location, one-sided values, corrected roots, $V_{\max}$, $P_{\min}$, benchmark statistics) was additionally reproduced by a second, independently written implementation of the map and CAPE routines, and the two agree to the digits shown. The #77 profile is the one committed in PR #77’s failing test; the 1995 profiles are those of `tests/test_oscillation_rescue.py`.

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